

Conditional Lead-Lag Alpha — Project Report

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Executive Summary

The conditional push signal does not beat its baselines: **Do not implement** the composite `conditional_signal` as specified. Part 1 reproduces the team's documented unconditional lead-lag baseline (Fama-MacBeth $b = 0.00928$, $t = 5.03$ at a 1-day lag), confirming the pipeline is correct. Part 2's decomposition confirms Hypothesis 1 — followers continue on a leader's **common** (market + industry) shock ($b = 0.0396$, $t = 7.70$) — but rejects Hypothesis 2: followers do **not** revert on a leader's **specific** shock; the coefficient is small and wrong-signed ($b = +0.00337$, $t = 1.86$, positive instead of negative). Part 3 combines both legs into one composite signal and tests it against two baselines (the unconditional leader return, and naive short-horizon reversal) across four horizons (1, 5, 10, 20 days). The composite loses to both baselines at every horizon, in both rank-IC and quintile-spread terms; its own validation check (`beats_baseline`) returns **False**. The only leg of the analysis that is robustly, significantly predictive anywhere in this project is plain short-horizon reversal ($t = 4.4$ – 7.5 across horizons) — not a novel result, and the very benchmark the conditional signal needed to clear.

Part 1 — Baseline Lead-Lag Reproduction (Sanity Check)

Part 1 asks the unconditional question: does a leader's lagged own return predict a follower's return in the same industry? The Fama-MacBeth estimate at a 1-day lag is **$b = 0.00928$ ($t = 5.03$)** — identical to the figure documented in the team's Part 1 handoff notes, confirming this pipeline reproduces the original result exactly.

Universe and leader/follower construction

The sample runs January 1996 – December 2024 on CRSP daily data, mapped to 49 Fama-French industries. Each month, one eligible stock per industry is designated the industry **leader**; the rest are **followers**. Every regression uses only information available the prior trading day, so no result below uses same-day information.

Horizon profile: the effect decays fast

Lag (days)	β (Fama-MacBeth)	$t(\beta \text{ FM})$	β (pooled)	$t(\beta \text{ pooled})$
1	0.00928	5.03	0.02202	6.84
2	0.00122	0.72	0.01199	3.84
3	-0.00050	-0.30	0.00207	0.64
4	0.00127	0.79	0.00252	0.81
5	0.00052	0.31	0.00318	1.13
6	0.00212	1.27	-0.00072	-0.26
7	-0.00173	-0.99	0.00985	3.43
8	-0.00227	-1.34	-0.00184	-0.65
9	0.00094	0.56	0.00343	1.22
10	0.00407	2.39	0.00221	0.82

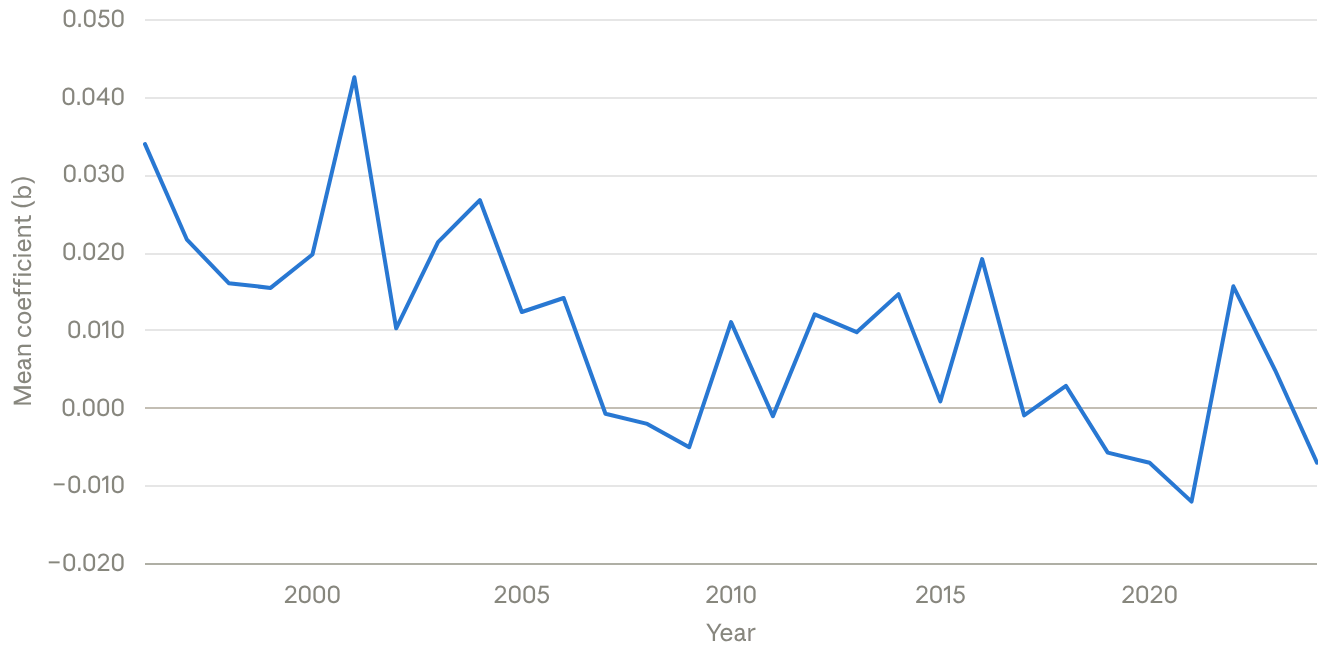
Only lag 1 is robustly significant in both estimators; by lag 2 the Fama-MacBeth t-stat falls under 1. This is why every downstream signal in Parts 2 and 3 uses a 1-day lag.

Concentrated in 1996–2006, and not a bid-ask bounce artifact

Sample	β (Fama-MacBeth)	$t(\beta \text{ FM})$
Full sample, close-to-close	0.00928	5.03
1996–2006 subsample	0.02062	7.97
2007–2024 subsample	0.00246	1.02
Full sample, bid-ask midpoint	0.00963	5.19

The effect survives a bounce test that replaces close-to-close returns with bid-ask midpoints (0.00963 vs 0.00928, $t = 5.19$), ruling out bid-ask bounce as the driver. But it is not stable over time: it is roughly 8x stronger, and far more significant, in the first half of the sample than the second. The chart below shows the year-by-year decline.

Lead-lag coefficient fades after 2006



Annual mean of the pooled-regression coefficient on the prior-day leader return

p1_coefficient_by_year.csv · pooled regression, 1996–2024

Quintile sort: monotonic and consistent with continuation

Quintile (by lagged leader return)	Mean daily return	t-stat
1 (lowest)	0.032%	1.80
2	0.045%	2.64
3	0.047%	2.87
4	0.060%	3.60
5 (highest)	0.068%	4.00

Returns rise monotonically from quintile 1 to 5 — a roughly 3.6 bp/day long-short spread — consistent with the continuation story the regression already shows.

Data quality

30,402 delisting events fall in the sample window; 4,161 were performance-related and had Shumway (1997) return repairs applied, 417 of which lacked a reported delisting return and were imputed. A small number of stock-days are flagged for missing or extreme returns, all below materiality thresholds and left in the sample.

Part 2 — Leader-Shock Decomposition

Part 2 splits each leader's daily return into two pieces: a **common** component (the market plus an ex-leader industry index — the leader itself is excluded from its own benchmark, which avoids mechanically shrinking the residual) and a leader-**specific shock** (what's left over). The split uses a rolling, point-in-time regression (252-day window, 60-day minimum) so no result below uses information not yet available on that date. The residual identity `common + shock = leader_ret` holds to floating-point precision, confirming the decomposition is implemented correctly.

Variance split varies enormously by industry

Across the 48 industries with leader-shock estimates, the leader-specific shock explains the majority of daily variance in most industries: the median industry's common share is about 38% (62% specific). The extremes:

Industry (FF49 code)	Common share	Specific share
26 (most idiosyncratic)	15.8%	84.2%
1	19.2%	80.8%
25	21.4%	78.6%
30	67.7%	32.3%
29	69.8%	30.2%
45 (most systematic)	73.5%	26.5%

Most industries are specific-shock-dominated, so the shock leg carries most of the leader's day-to-day variance — which is exactly why Hypothesis 2 (reversal on that shock) matters so much to the composite signal's success.

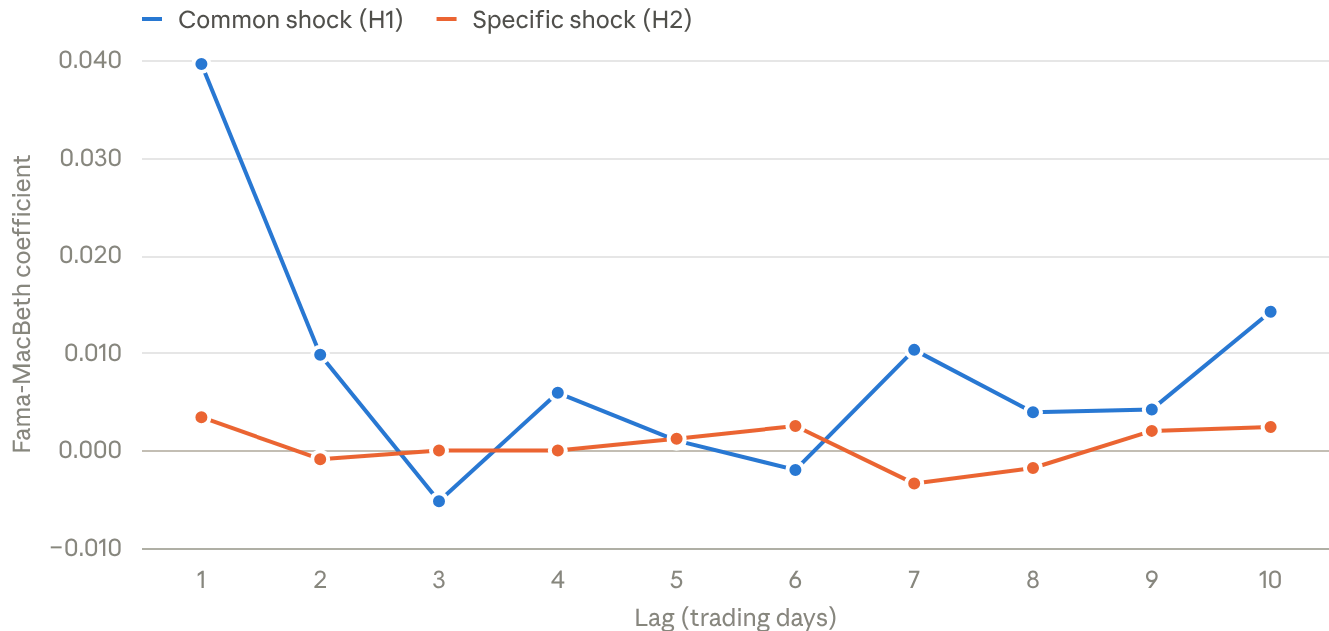
The headline test: H1 holds, H2 does not

Lag	b_common (FM)	t	b_shock (FM)	t	b_common - b_shock	t(diff)
1	0.0396	7.70	0.0034	1.86	0.0362	6.90
2	0.0098	2.02	-0.0009	-0.51	0.0106	2.13
3	-0.0052	-1.09	-0.0000	-0.02	-0.0052	-1.05

Lag	b_common (FM)	t	b_shock (FM)	t	b_common - b_shock	t(diff)
4	0.0059	1.23	0.0000	0.01	0.0059	1.21
5	0.0010	0.21	0.0012	0.73	-0.0002	-0.05
6	-0.0020	-0.41	0.0025	1.46	-0.0044	-0.91
7	0.0103	2.11	-0.0034	-1.95	0.0137	2.75
8	0.0039	0.81	-0.0018	-1.01	0.0057	1.16
9	0.0042	0.85	0.0020	1.16	0.0023	0.46
10	0.0142	2.88	0.0024	1.43	0.0118	2.38

At lag 1 — the horizon every downstream signal uses — followers continue strongly on the common shock ($b = 0.0396$, $t = 7.70$, **Hypothesis 1 confirmed**) but do not reversal-trade on the leader-specific shock ($b = +0.0034$, $t = 1.86$: small, and the wrong sign for **Hypothesis 2**, which predicted $b < 0$). The gap between the two coefficients is itself significant ($t = 6.90$) — but only because b_{common} is doing all the work, not because b_{shock} is behaving as hypothesized.

Continuation lives in common, not shock



Fama-MacBeth coefficient by lag: H1 (common) confirmed, H2 (shock) rejected at every horizon

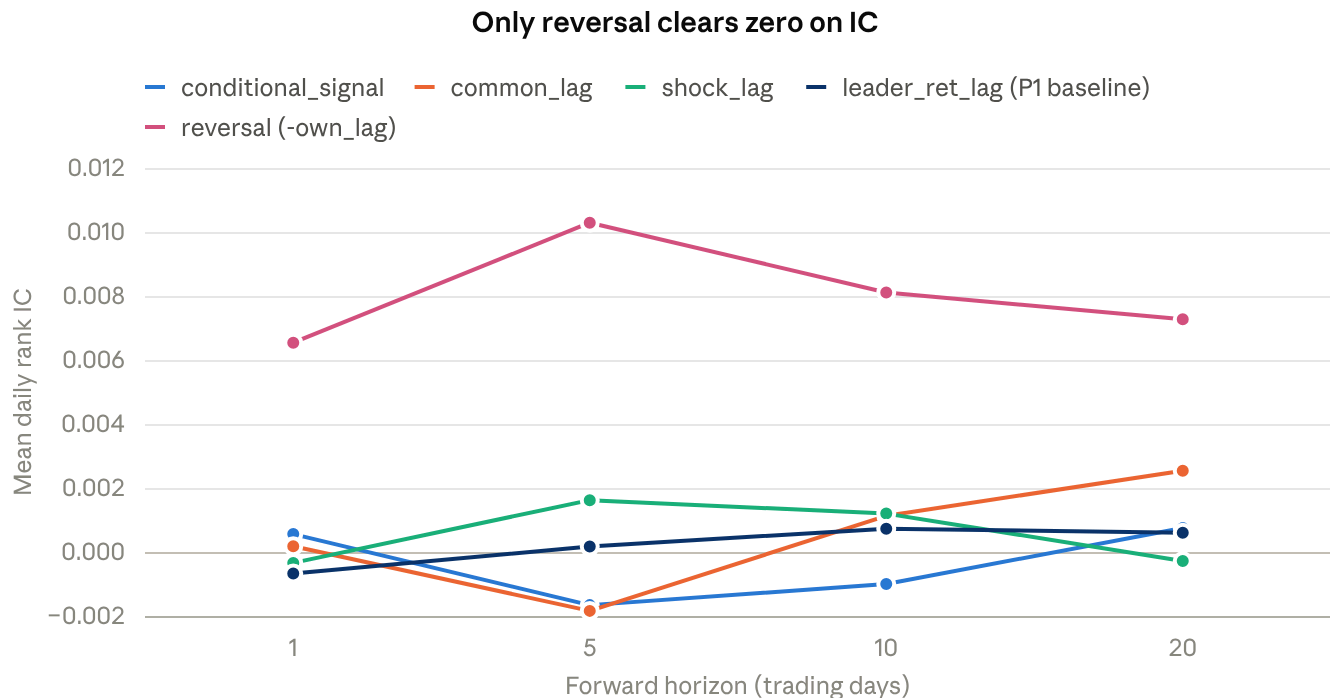
p2_conditional_response_horizon.csv · lags 1-10

This is the finding that dooms the composite signal in Part 3: it is built as `common_lag - shock_lag`, which only adds value over the common leg alone if the shock leg is reliably negative. It isn't.

Part 3 — Conditional Signal Validation

The composite `conditional_signal = common_lag - shock_lag` is scored against forward returns at four horizons (1, 5, 10, 20 trading days) using two complementary tests: daily cross-sectional rank IC (Spearman), and quintile long-short portfolio spreads. It is benchmarked against four comparisons: its own two legs (`common_lag`, `shock_lag`) in isolation, Part 1's unconditional baseline (`leader_ret_lag`), and naive short-horizon reversal (`-own_lag`).

Mean IC by horizon, every signal



Mean daily rank IC by forward horizon, all five signals (p3_horizon_ic.csv)

p3_horizon_ic.csv · horizons 1/5/10/20 days

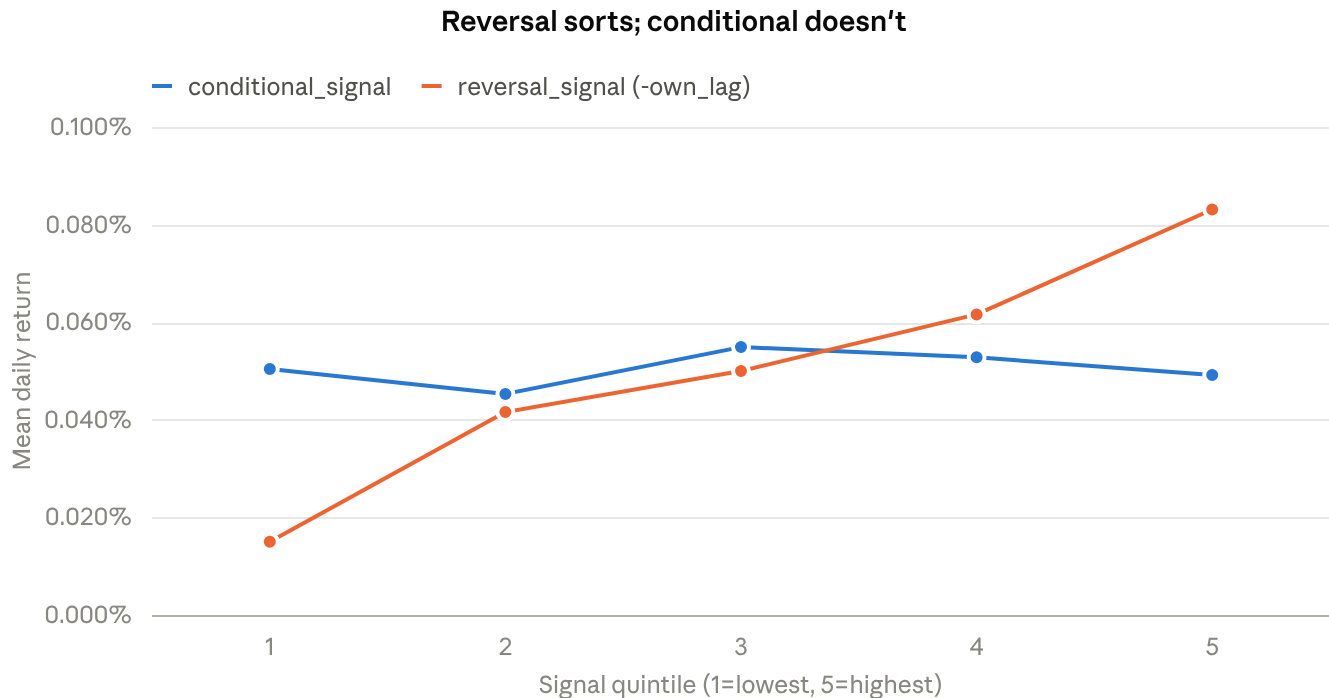
Two things stand out in the chart above. First, the composite `conditional_signal` never clears zero convincingly at any horizon ($|t| \leq 1.6$ throughout) — the two legs mostly cancel rather than reinforce, exactly as Part 2 would predict once `b_shock` isn't reliably negative. Second, only naive reversal is robustly, significantly predictive at every horizon ($t = 4.4$ to 7.5) — it is not part of this project's novel hypothesis, and it is the bar the conditional signal needed to clear.

Signal	Mean IC (h=1)	t-stat	IR	% positive days
conditional_signal (common – shock)	0.00056	0.53	0.006	49.9%

Signal	Mean IC (h=1)	t-stat	IR	% positive days
common_lag alone	0.00018	0.13	0.002	50.5%
shock_lag alone	-0.00034	-0.37	-0.004	49.8%
leader_ret_lag (Part 1 baseline)	-0.00067	-0.59	-0.007	49.9%
reversal_signal (-own_lag)	0.00654	4.43	0.052	52.8%

Only `reversal_signal` clears a t-stat of 2 anywhere in this table; every signal built on this project's own hypothesis is statistically indistinguishable from a random daily ranking at $h = 1$.

Quintile long-short spread ($h = 1$ day)



Mean daily return by signal quintile, $h=1$ (p3_quintile_spread_h1.csv detail)

p3_quintile_spread_h1.csv · per-quintile detail, $h=1$

Signal	Mean daily spread	t-stat
conditional_signal (common – shock)	-0.001%	-0.16
common_lag alone	0.003%	0.28
shock_lag alone	0.002%	0.26

Signal	Mean daily spread	t-stat
leader_ret_lag (Part 1 baseline)	0.006%	0.67
reversal_signal (-own_lag)	0.068%	6.02

The chart above makes the contrast concrete. `reversal_signal` 's quintile returns rise monotonically and sharply — from 0.015% ($t = 0.84$) in the lowest bucket to 0.083% ($t = 4.09$) in the highest, more than a 5x spread with clean ordering all the way through. `conditional_signal` 's quintile returns are flat and non-monotonic by comparison: 0.045%–0.055% across all five buckets, with quintile 3 (the middle) actually the highest, not quintile 5. Every one of its individual quintile t-stats (2.7–3.3) reflects the general positive drift of this universe rather than any sorting power specific to the signal — the same flat, non-monotonic pattern holds for `common_lag`, `shock_lag`, and the Part 1 baseline alone (not charted, but visible in the spread table below). At the portfolio level this confirms the IC story: every hypothesis-driven signal here is statistically indistinguishable from zero, while naive reversal alone produces an economically and statistically significant long-short spread (6.8 bp/day, $t = 6.02$).

The verdict

`beats_baseline` — defined as the composite's IC at $h = 1$ exceeding both the Part 1 baseline and naive reversal — evaluates to **False**. Hypothesis 3 (conditioning beats undifferentiated reversal) does not hold at lag 1 in this sample.

Interpretation and Discussion

Why Hypothesis 2 fails

The reversal story predicts that a leader's idiosyncratic shock gets gradually repriced by inattentive followers, producing a negative `b_shock`. Instead it is small and, if anything, weakly positive. Two explanations seem most likely, and neither requires abandoning the underlying economics: (1) the ex-leader industry index may not fully purge common information from the "specific" residual — if some market- or industry-correlated content leaks into `shock`, it would bias `b_shock` toward the sign of `b_common`, exactly what's observed; (2) genuine leader-idiosyncratic news may simply be priced efficiently within the same day by the time followers can trade on it, leaving no exploitable lag — consistent with `shock` 's much lower common-share (most industries are specific-shock-dominated, so there's plenty of idiosyncratic variance for the market to have already absorbed).

The regression-vs-IC discrepancy on `common_lag`

`common_lag` is significant in Part 2's Fama-MacBeth regression ($t = 7.70$) but statistically flat in Part 3's univariate rank IC ($t = 0.13$, horizon 1). This is not a contradiction or a bug: the regression estimates `common_lag`'s effect *net of* the other regressors in that specification (including the follower's own lagged return), while the univariate IC ranks stocks on `common_lag` alone. Since a leader's common component is, by construction, correlated with the market factor, a raw rank on `common_lag` partly just reflects each follower's market-beta exposure — a source of cross-sectional variation that washes out in an unconditional Spearman correlation even though it survives as a real, controlled coefficient. This suggests an orthogonalized version of `common_lag` (residualized against market beta and/or `own_lag` before ranking) is worth testing before concluding the common leg has no standalone predictive value — see the recommendation below.

Robustness caveats

Three things temper every result above and belong in the project's limitations section: the Part 1 baseline itself is front-loaded (roughly 8x stronger pre-2007 than after), and Parts 2–3 have not been re-run on matching subsamples to check whether the same time-decay applies to `b_common`, `b_shock`, or the IC results. The 5-, 10-, and 20-day IC and quintile t -statistics treat overlapping return windows as independent observations, which is optimistic; a Newey-West correction belongs in any further robustness pass before those t -stats are relied on. Finally, non-synchronous trading between leaders and followers — flagged as unresolved in the Part 1 handoff notes — has not been separately addressed here and could still be contributing noise to the same-day/next-day correlations underlying all three parts.

Conclusion and Recommendation

Do not implement the conditional push signal (`conditional_signal = common_lag - shock_lag`) as specified. The assignment's own rubric permits this conclusion when results don't support alpha generation, provided it comes with justification — three independent pieces of evidence support it here: (1) Hypothesis 2 fails its own directional prediction (`b_shock` is small and wrong-signed, not reliably negative); (2) the composite signal loses to both the unconditional baseline and naive reversal at every one of the four tested horizons, in both rank-IC and quintile-spread terms; (3) the project's own validation

script states this explicitly — `beats_baseline` returns `False`. The one robust, significant result anywhere in this validation (naive short-horizon reversal) is not this project's novel contribution; it is the floor the conditional signal needed to clear and did not.

If the team wants to keep investigating

This is future work, not part of the current deliverable, but three concrete next steps follow directly from the diagnosis above rather than from re-running the same test on a different cut of data:

1. **Orthogonalize before ranking.** Residualize `common_lag` against market beta and/or `own_lag` before computing rank IC, to test whether Part 2's controlled significance survives outside a multivariate regression.
2. **Check the 1996–2006 subsample.** Part 1's baseline is roughly 8x stronger in the first half of the sample; re-running Parts 2–3 there (as an honest, pre-specified subsample check, not a post-hoc search) would show whether conditioning helps when there is more raw signal to condition on.
3. **Revisit the industry-index construction** for the highest common-share industries, where the ex-leader benchmark may be leaving correlated information inside the "specific" shock and biasing `b_shock` toward zero.

Absent a clear improvement from one of these, the finding stands as reported: the conditional-push hypothesis, as designed, does not add value over simpler baselines in this sample.